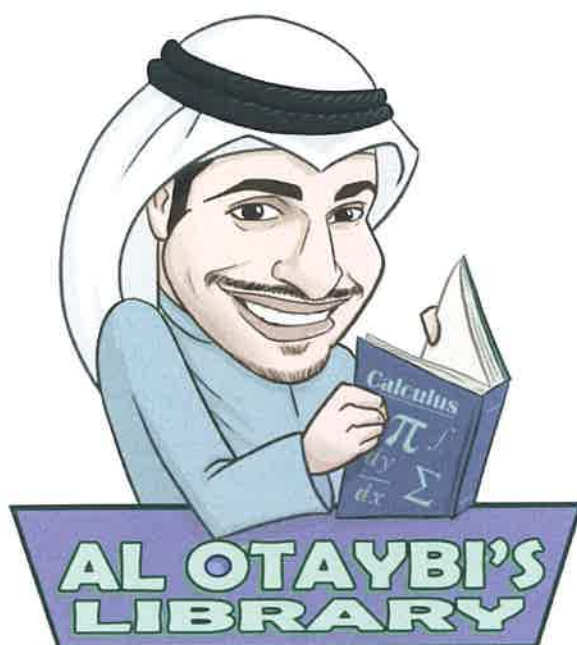


Kuwait University

2021/2022

Fall course



Calculus A

Second

Section : 4.2

The Mean Value Theorem.

(14)



4.2 The Mean Value Theorem



Rolle's Theorem (R.T)



If :

- 1) f is continuous on $[a, b]$.
- 2) f is differentiable on (a, b)
- 3) $f(a) = f(b)$

Then there is a number c in (a, b) such that :

$$f'(c) = 0$$



Mean Value Theorem (M.V.T)



If :

- 1) f is continuous on $[a, b]$.
- 2) f is differentiable on (a, b)

Then there is a number c in (a, b) such that :

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$



Text book question



Verify that the function satisfies the hypotheses of Rolle's Theorem on the given interval. Then find all numbers c that satisfy the conclusion of Rolle's Theorem .

$$f(x) = 2x^2 - 4x + 5, [-1, 3]$$

$$f(x) = 2x^2 - 4x + 5, [-1, 3]$$

1) Domain of $f(x) = \mathbb{R}$

$\therefore f(x)$ is continuous on $[-1, 3]$

2) $f'(x) = 4x - 4$

$f(x)$ is differentiable on $\mathbb{R}$

$\therefore f(x)$ is differentiable on $(-1, 3)$

3) $f(-1) = 2 + 4 + 5 = 11$

$$f(3) = 2(9) - 4(3) + 5 = 18 - 12 + 5 = 11$$

$$\therefore f(-1) = f(3)$$

From ① ② ③ $\longrightarrow \therefore$ Rolle's Theorem is satisfied
then there is a number $c \in (-1, 3)$ such that

$$f'(c) = 0$$

$$4c - 4 = 0$$

$$4c = 4 \xrightarrow{\div 4} \boxed{c = 1} \in (-1, 3)$$



Text book question



Verify that the function satisfies the hypotheses of the Mean Value Theorem on the given interval. Then find all numbers c that satisfy the conclusion of the Mean Value Theorem .

$$f(x) = x^3 - 3x + 2, \quad [-2, 2]$$

$$f(x) = x^3 - 3x + 2 \quad [-2, 2]$$

- 1) Domain of $f(x) = \mathbb{R}$
 $\therefore f(x)$ is continuous on $[-2, 2]$
- 2) $f'(x) = 3x^2 - 3$
 $f(x)$ is differentiable on $\mathbb{R}$
 $\therefore f(x)$ is differentiable on $(-2, 2)$

From ① ② $\longrightarrow \therefore$ Mean Value Theorem is Satisfied
then there is a number $c \in (-2, 2)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$3c^2 - 3 = \frac{f(2) - f(-2)}{2 - (-2)}$$

$$3c^2 - 3 = \frac{(8 - 6 + 2) - (-8 + 6 + 2)}{4}$$

$$3c^2 - 3 = \frac{4 - 0}{4}$$

$$3c^2 - 3 = 1$$

$$3c^2 - 3 = 1$$

$$3c^2 = 4$$

$$c^2 = \frac{4}{3}$$

$$c = \frac{2}{\sqrt{3}}$$

$$\in (-2, 2)$$

$$c = -\frac{2}{\sqrt{3}}$$

$$\in (-2, 2)$$

Check if $f(x)$ satisfies Mean Value Theorem on the given interval . If yes , find c

$$f(x) = \sqrt[3]{x^2} , [-8, 8]$$

$$f(x) = x^{\frac{2}{3}} \quad [-8, 8]$$

1) Domain of $f(x) = \mathbb{R}$

$\therefore f(x)$ is continuous on $[-8, 8]$

$$2) f'(x) = \frac{2}{3} x^{-\frac{1}{3}} = \frac{2}{3 x^{\frac{1}{3}}}$$

$f(x)$ is differentiable on $\mathbb{R} \setminus \{0\}$

$$\therefore 0 \in (-8, 8)$$

$\therefore f(x)$ is not differentiable on $(-8, 8)$

$\therefore$ Mean Value Theorem is not satisfied

$\therefore$ No value of $c \in (-8, 8)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$f'(c) = \frac{f(8) - f(-8)}{8 - (-8)}$$

Check if $f(x)$ satisfies Mean Value Theorem on the given interval. If yes, find c

$$f(x) = \sqrt[3]{x^2}, [1, 8]$$

$$f(x) = x^{\frac{2}{3}} \quad [1, 8]$$

1) Domain of $f(x) = \mathbb{R}$

$\therefore f(x)$ is continuous on $[1, 8]$

$$2) f'(x) = \frac{2}{3} x^{-\frac{1}{3}} = \frac{2}{3x^{\frac{1}{3}}}$$

f is differentiable on $\mathbb{R} \setminus \{0\}$

$\therefore 0 \notin (1, 8)$

$\therefore f$ is differentiable on $(1, 8)$

From ①② $\longrightarrow \therefore$ Mean Value Theorem is satisfied
then there is a number $c \in (1, 8)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\frac{2}{3c^{\frac{1}{3}}} = \frac{f(8) - f(1)}{8 - 1}$$

$$\frac{2}{3c^{\frac{1}{3}}} = \frac{(8)^{\frac{2}{3}} - (1)^{\frac{2}{3}}}{7}$$

$$\frac{2}{3c^{\frac{1}{3}}} = \frac{(2^3)^{\frac{2}{3}} - 1}{7}$$

$$\frac{2}{3c^{\frac{1}{3}}} = \frac{4 - 1}{7}$$

$$\frac{2}{3c^{\frac{1}{3}}} = \frac{3}{7}$$

$$14 = 9c^{\frac{1}{3}}$$

$$c^{\frac{1}{3}} = \frac{14}{9}$$

$$c = \left(\frac{14}{9}\right)^3 \in (1, 8)$$

Verify that the function $f(x) = x + \frac{1}{x}$ satisfies the hypotheses of Rolle's Theorem on the interval $[\frac{1}{2}, 2]$.

Then find all numbers c , if any, that satisfy the conclusion of Rolle's Theorem

Apr 2019

$$f(x) = x + \frac{1}{x} \quad [\frac{1}{2}, 2]$$

1) Domain of $f(x)$: $\mathbb{R} \setminus \{0\}$

$$\because 0 \notin [\frac{1}{2}, 2]$$

$\therefore f$ is continuous on $[\frac{1}{2}, 2]$

$$2) f'(x) = 1 - \frac{1}{x^2}$$

f is differentiable on $\mathbb{R} \setminus \{0\}$

$$\because 0 \notin (\frac{1}{2}, 2)$$

$\therefore f$ is differentiable on $(\frac{1}{2}, 2)$

$$3) f(\frac{1}{2}) = \frac{1}{2} + \frac{1}{\frac{1}{2}} = \frac{1}{2} + 2 = 2.5$$

$$f(2) = 2 + \frac{1}{2} = 2.5$$

$$\therefore f(\frac{1}{2}) = f(2)$$

From ① ② ③ $\longrightarrow \therefore$ Rolle's Theorem is satisfied

then there is a number $c \in (\frac{1}{2}, 2)$ such that

$$\begin{aligned} f'(c) &= 0 \\ 1 - \frac{1}{c^2} &= 0 \\ 1 &= \frac{1}{c^2} \end{aligned} \quad \begin{aligned} c^2 &= 1 \\ c &= -1 \\ &\notin (\frac{1}{2}, 2) \end{aligned}$$

$$\begin{aligned} c &= 1 \\ &\in (\frac{1}{2}, 2) \end{aligned}$$

Let $f(x) = \ln(x^2 + 1)$.

Dec 2016

- a) Verify that f satisfies the condition of Rolle's Theorem on $[-1, 1]$
- b) Find all $c \in (-1, 1)$ that satisfy the conclusion of Rolle's Theorem.

$$f(x) = \ln(x^2 + 1) \quad [-1, 1]$$

1) Domain of $f(x)$: $x^2 + 1 > 0 \rightarrow x^2 > -1$
 $\Rightarrow D_f = \mathbb{R}$

$\therefore f(x)$ is continuous on $[-1, 1]$

2) $f'(x) = \frac{2x}{x^2 + 1}$
 $x^2 + 1 = 0 \rightarrow x^2 = -1$ None

$\therefore f$ is differentiable on $\mathbb{R}$

$\therefore f$ is differentiable on $(-1, 1)$

3) $f(-1) = \ln(1+1) = \ln 2$
 $f(1) = \ln(1+1) = \ln 2 \rightarrow \therefore f(-1) = f(1)$

From ①②③ $\rightarrow \therefore$ Rolle's Theorem is satisfied
then there is a number $c \in (-1, 1)$ such that

$$f'(c) = 0$$

$$\frac{2c}{c^2 + 1} = 0$$

* $(c^2 + 1)$

$$2c = 0$$

$$\div 2$$

$$c = 0$$

$$c \in (-1, 1)$$

Verify that $f(x) = \sqrt{4 - x^2}$ satisfies the hypotheses of the Mean Value Theorem on the interval $[-2, 0]$. Then, find all numbers c that satisfy the conclusion of the Theorem.

July 2017

1) Domain of $f(x)$: $4 - x^2 \geq 0$
 $4 \geq x^2$

$$\Rightarrow \therefore D_f = [-2, 2]$$

$\therefore f$ is continuous on $[-2, 0]$

2) $f'(x) = \frac{-2x}{2\sqrt{4-x^2}} = \frac{-x}{\sqrt{4-x^2}}$

f is differentiable on $(-2, 2)$

$\therefore f$ is differentiable on $(-2, 0)$

From ①② $\longrightarrow \therefore$ Mean Value Theorem is satisfied
 then there is a number $c \in (-2, 0)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a} \longrightarrow \frac{-c}{\sqrt{4-c^2}} = 1$$

$$\frac{-c}{\sqrt{4-c^2}} = \frac{f(0) - f(-2)}{0 - (-2)}$$

$$\frac{-c}{\sqrt{4-c^2}} = \frac{\sqrt{4-0} - \sqrt{0}}{2}$$

$$\frac{-c}{\sqrt{4-c^2}} = \frac{2}{2}$$

$$-c = \sqrt{4-c^2}$$

$$c^2 = 4 - c^2$$

$$2c^2 = 4$$

$$\div 2 \quad c^2 = 2$$

$$c = \sqrt{2}$$

$$c = -\sqrt{2} \in (-2, 0)$$

if $f(x) = \sqrt[3]{x-1}$.

Use Mean Value Theorem to show that there is no value of $c \in (0, 4)$ such that $f(4) - f(0) = 4f'(c)$

$f(x) = (x-1)^{\frac{1}{3}} \quad [0, 4]$

1) Domain of $f(x) = \mathbb{R}$

$\therefore f(x)$ is continuous on $[0, 4]$

2) $f'(x) = \frac{1}{3}(x-1)^{-\frac{2}{3}} = \frac{1}{3(x-1)^{\frac{2}{3}}}$
 $f(x)$ is differentiable on $\mathbb{R} \setminus \{1\}$

$\therefore 1 \in (0, 4)$

$\therefore f(x)$ is not differentiable on $(0, 4)$

$\therefore$ Mean Value Theorem is not satisfied

$\therefore$ No value of $c \in (0, 4)$ such that :

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$f'(c) = \frac{f(4) - f(0)}{4 - 0}$$

$\times 4$

$$f(4) - f(0) = 4f'(c)$$



Text book question



Let $f(x) = (x - 3)^{-2}$. Show that there is no value of c in $(1, 4)$ such that $f(4) - f(1) = f'(c)(4 - 1)$.

Why does this not contradict the Mean Value Theorem.

$$f(x) = \frac{1}{(x-3)^2} = (x-3)^{-2}$$

$$f(4) = \frac{1}{(4-3)^2} = \frac{1}{(1)^2} = 1$$

$$f(1) = \frac{1}{(1-3)^2} = \frac{1}{(-2)^2} = \frac{1}{4}$$

$$\Rightarrow f(4) - f(1) = 1 - \frac{1}{4} = \frac{4}{4} - \frac{1}{4} = \frac{3}{4}$$

$$f'(x) = -2(x-3)^{-3} = \frac{-2}{(x-3)^3}$$

$$\Rightarrow f'(c)(4-1)$$

$$\frac{-2}{(c-3)^3} \cdot 3 = \frac{-6}{(c-3)^3}$$

$$f(4) - f(1) = f'(c)(4-1)$$

$$\frac{3}{4} = \frac{-6}{(c-3)^3}$$

$$3(c-3)^3 = -24$$

$$\div (-3)$$

$$(c-3)^3 = -8$$

$$c-3 = -2$$

$$c = 1 \notin (1, 4)$$

* No value of c such that $f(4) - f(1) = f'(c)(4-1)$

$$f(x) = \frac{1}{(x-3)^2} = (x-3)^{-2} \quad [1,4]$$

1) Domain of $f(x) = \mathbb{R} \setminus \{3\}$

$$\therefore 3 \in [1,4]$$

$\therefore f(x)$ is not continuous on $[1,4]$

$\therefore$ Mean Value Theorem is not satisfied

$\therefore$ No value of $c \in (1,4)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b-a}$$

$$f'(c) = \frac{f(4) - f(1)}{4-1}$$

$$(4-1) f'(c) = f(4) - f(1)$$



Let $f(x) = |x - 1|$

(a) Show that there is no $c \in (0, 3)$ such that

$$f'(c) = \frac{1}{3}(f(3) - f(0))$$

(b) Explain why this result in part (a) does not contradict the Mean value theorem?

$$a) \quad f(x) = \begin{cases} x-1, & x \geq 1 \\ -(x-1), & x < 1 \end{cases}$$

$$f'(x) = \begin{cases} 1, & x > 1 \\ -1, & x < 1 \end{cases}$$

$$\Rightarrow f'(c) = \begin{cases} 1, & c > 1 \\ -1, & c < 1 \end{cases}$$

$$f(3) = 3 - 1 = 2$$

$$f(0) = 1 - 1 = 0$$

$$\Rightarrow \frac{1}{3}(f(3) - f(0)) = \frac{1}{3}(2 - 0) = \frac{2}{3}$$

$$\therefore f'(c) \neq \frac{1}{3}(f(3) - f(0))$$

$\therefore$ No value of c such that

$$f'(c) = \frac{1}{3}(f(3) - f(0))$$

(b) $f(x) = |x-1|$

① $f(x)$ is continuous on $\mathbb{R}$ [absolute value]

$\therefore f$ is continuous on $[0, 3]$

② $f(x)$ is differentiable on $\mathbb{R} \setminus \{1\}$

$\therefore 1 \in (0, 3)$

$\therefore f$ is Not differentiable on $(0, 3)$

$\therefore$ Mean Value Thm is Not satisfied

$\Rightarrow$ the No c such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$f'(c) = \frac{f(3) - f(0)}{3 - 0}$$

$$f'(c) = \frac{1}{3} (f(3) - f(0))$$

$$\text{Let } f(x) = \begin{cases} x & , \text{ if } 0 \leq x < 1 \\ 2 - x & , \text{ if } 1 \leq x \leq 2 \end{cases}$$

(a) Show that $f(2) = f(0)$

(b) Show that $f'(c) \neq 0$ for all $c \in (0, 2)$

(c) Does this not contradict the Rolle's theorem? Explain.

a) $f(2) = 2 - 2 = 0$

$f(0) = 0$

$\therefore f(2) = f(0)$

b) $f'(x) = \begin{cases} 1 & , 0 < x < 1 \\ -1 & , 1 < x < 2 \end{cases}$

$f'(c) = \begin{cases} 1 & , 0 < c < 1 \\ -1 & , 1 < c < 2 \end{cases}$

$\therefore f'(c) \neq 0 \longrightarrow \therefore \text{No value of } c \in (0, 2)$
 $f'(c) = 0$

c) 1) $f(x)$ is continuous on $[0, 2]$
 2) $f'(\bar{1}) = 1, f'(\bar{1}^+) = -1$

$\therefore f'(\bar{1}) \neq f'(\bar{1}^+) \longrightarrow \therefore f$ is not differentiable
 at $x = 1$

$\therefore 1 \notin (0, 2)$

$\therefore f(x)$ is not differentiable on $(0, 2)$

$\therefore$ Rolle's Theorem is not satisfied

$\therefore$ No value of $c \in (0, 2)$ such that $f'(c) = 0$



Tricky question !!



Let $f(x) = 3x + \frac{9}{5}(x-1)^{\frac{5}{3}} + 1$

a) Show that there is no $c \in (0, 2)$ such that $f''(c) = 0$.

b) Explain why this result does not contradict Rolle's Theorem.

$$\begin{aligned} \text{a) } f'(x) &= 3 + \frac{9}{5} \cdot \frac{5}{3} (x-1)^{\frac{2}{3}} \quad (1) \\ &= 3 + 3(x-1)^{\frac{2}{3}} \end{aligned}$$

$$f''(x) = 0 + 3 \cdot \frac{2}{3} (x-1)^{-\frac{1}{3}} = \frac{2}{(x-1)^{\frac{1}{3}}}$$

$$\Rightarrow f''(c) = \frac{2}{(c-1)^{\frac{1}{3}}}$$

$$\frac{2}{(c-1)^{\frac{1}{3}}} = 0$$

$$\times (c-1)^{\frac{1}{3}} \rightarrow 2 = 0 \quad (\text{which is wrong})$$

$\therefore$ No value of c such that $f''(c) = 0$

b) $f(x) = 3x + \frac{9}{5}(x-1)^{\frac{5}{3}} + 1$

$$f'(x) = 3 + \frac{9}{5} \cdot \frac{5}{3} (x-1)^{\frac{2}{3}}$$

$$\begin{aligned} \text{Let } h(x) &= f'(x) \\ h(x) &= 3 + 3(x-1)^{\frac{2}{3}} \end{aligned}$$

1) Domain of $h(x) = \mathbb{R} \rightarrow \therefore h(x)$ is continuous on $[0, 2]$

$$2) h'(x) = 0 + 3 \cdot \frac{2}{3} (x-1)^{-\frac{1}{3}} = \frac{2}{(x-1)^{\frac{1}{3}}}$$

h is differentiable on $\mathbb{R} \setminus \{1\}$

$\therefore 1 \in (0, 2) \rightarrow \therefore h$ is Not differentiable on $(0, 2)$

$\therefore$ Rolle's Theorem is not satisfied

$\therefore$ No value of c such that $h(c) = 0$
 $f''(c) = 0$

Let $f'(x) = 0$ for all values of x

Show that $f(a) = f(b)$ for any real number a and b

(Use Mean value theorem)

$f'(x) = 0$ $[a, b]$

$\therefore f'(x) = 0 \quad \forall x$

- $\therefore$ ① $f(x)$ is continuous on $[a, b]$
- $\therefore$ ② $f(x)$ is differentiable on (a, b)

From ① ② $\implies \therefore$ Mean Value Theorem is satisfied
then there is a number $c \in (a, b)$

such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$0 = f(b) - f(a)$$

$$\boxed{f(a) = f(b)}$$

Let $h(x)$ be differentiable function satisfying

$h(15) = 123$, $h'(x) \leq 10$ for all real number x .

Show that $h(6) \geq 33$

$h'(x) \leq 10$

$[6, 15]$

$\therefore h'(x) \leq 10$ $\forall x$ $\begin{cases} \therefore 1) h(x) \text{ is differentiable on } (6, 15) \\ \therefore 2) h(x) \text{ is continuous on } [6, 15] \end{cases}$

From ① ② $\implies \therefore$ Mean Value Theorem is satisfied
then there is a number $c \in (6, 15)$
such that

$$h'(c) = \frac{h(b) - h(a)}{b - a}$$

$$10 \geq \frac{h(15) - h(6)}{15 - 6}$$

$$10 \geq \frac{123 - h(6)}{9}$$

$$90 \geq 123 - h(6)$$

$$h(6) \geq 123 - 90$$

$$h(6) \geq 33$$

Use mean value theorem to show that if f is differentiable on R , $f(0) = 1$ and $f'(x) \geq 2$ for all $x > 0$ then $f(x) \geq 2x + 1$ for all $x > 0$

$$f'(x) \geq 2$$

$$[0, x]$$

$$\therefore f'(x) \geq 2 \quad \forall x \begin{cases} \Rightarrow \textcircled{1} f(x) \text{ is continuous on } [0, x] \\ \Rightarrow \textcircled{2} f(x) \text{ is differentiable on } (0, x) \end{cases}$$

From $\textcircled{1} \textcircled{2} \Rightarrow \therefore$ Mean Value Theorem is satisfied
 then there is a number $c \in (0, x)$
 such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$2 \leq \frac{f(x) - f(0)}{x - 0}$$

$$2 \leq \frac{f(x) - 1}{x}$$

* x

$$2x \leq f(x) - 1$$

$$2x + 1 \leq f(x)$$

$$f(x) \geq 2x + 1$$

Let f be a function such that $f'(x) < k$ for all x with $f(0) = 3$ and $f(3) = 6$

Find all possible values of k for such function to exist

$$f'(x) < k$$

$$[0, 3]$$

$$\therefore f'(x) < k \begin{cases} \nearrow \therefore \textcircled{1} f(x) \text{ is continuous on } [0, 3] \\ \searrow \forall x \therefore \textcircled{2} f(x) \text{ is differentiable on } (0, 3) \end{cases}$$

From $\textcircled{1} \textcircled{2} \Rightarrow \therefore$ Mean Value Theorem is satisfied
then there is a number $c \in (0, 3)$
such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$k > \frac{f(3) - f(0)}{3 - 0}$$

$$k > \frac{6 - 3}{3}$$

$$k > \frac{3}{3}$$

$$k > 1$$

$$\Rightarrow \therefore k \in (1, \infty)$$



Text book question



Suppose that $3 \leq f'(x) \leq 5$ for all values of x .

Show that $18 \leq f(8) - f(2) \leq 30$

$$3 \leq f'(x) \leq 5$$

$$[2, 8]$$

$\therefore 3 \leq f'(x) \leq 5$ for all x

- 1) $\therefore f$ is continuous on $[2, 8]$
- 2) $\therefore f$ is differentiable on $(2, 8)$

From ①② $\longrightarrow \therefore$ Mean Value Theorem is satisfied

Then there is a number $c \in (2, 8)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$3 \leq \frac{f(8) - f(2)}{8 - 2} \leq 5$$

$$3 \leq \frac{f(8) - f(2)}{6} \leq 5$$

$$\times 6$$

$$18 \leq f(8) - f(2) \leq 30$$



Text book question



Does there exist a function f such that

$f(0) = -1$, $f(2) = 4$ and $f'(x) \leq 2$ for all x ?

$$f'(x) \leq 2$$

$$[0, 2]$$

$$\therefore f'(x) \leq 2 \text{ for all } x$$

1) $\therefore f(x)$ is continuous on $[0, 2]$

2) $\therefore f(x)$ is differentiable on $(0, 2)$

From ① ② $\longrightarrow \therefore$ Mean Value Theorem is satisfied

Then there is a number $c \in (0, 2)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$2 \nparallel \frac{f(2) - f(0)}{2 - 0}$$

$$2 \nparallel \frac{4 - (-1)}{2}$$

*2

4 $\nparallel$ 5 (which is wrong)

$\therefore f(x)$ D.N.E

Given a function $f(x)$ such that $f(6) = 5$ and $f'(x) \leq -5$ for all $x \in [6, 10]$. How large can $f(10)$ possible be ?

Dec 2017

$$f'(x) \leq -5, [6, 10]$$

$\therefore f'(x) \leq -5$ for all $x \in [6, 10]$

- 1) $\therefore f$ is continuous on $[6, 10]$
- 2) $\therefore f$ is differentiable on $(6, 10)$

From ① ② $\longrightarrow \therefore$ Mean Value Theorem is satisfied then there is a number $c \in (6, 10)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$-5 \leq \frac{f(10) - f(6)}{10 - 6}$$

$$\times 4 \quad -5 \leq \frac{f(10) - 5}{4}$$

$$-20 \leq f(10) - 5$$

$$-15 \leq f(10) \longrightarrow \therefore \text{The largest of possible value of } f(10) = -15$$



Homeork !



Suppose that $f(0) = -3$ and $f'(x) \leq 5 \forall x$

How large can $f(2)$ possible be ?

(Final Answer : 7)

Suppose f is defined on $(-\infty, \infty)$,

$$f'(x) = 1 \quad \forall x, \quad f(2) = 3$$

Use Mean Value Theorem to show that $f(x) = x + 1 \quad \forall x$

~~$f(x) = x + 1$~~

~~$(-\infty, \infty)$~~

$f'(x) = 1$

$[2, x]$

$\therefore f'(x) = 1$
 $\forall x$

1) $\therefore f(x)$ is continuous on $[2, x]$

2) $\therefore f(x)$ is differentiable on $(2, x)$

From ① ② $\longrightarrow \therefore$ Mean Value Theorem is satisfied
then there is a number $c \in (2, x)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$1 = \frac{f(x) - f(2)}{x - 2}$$

$$1 = \frac{f(x) - 3}{x - 2}$$

$\times (x - 2)$

$$x - 2 = f(x) - 3$$

$$\therefore f(x) = x + 1$$

$$f(a) - g(a) = 0$$

لنقارن

4) let f, g be continuous on $[a, b]$, and differentiable on (a, b)
Use Rolle's Theorem to show that if $f(a) = g(a)$ and $f(b) = g(b)$
then there is a point $c \in (a, b)$ such that $f'(c) = g'(c)$

لو كان عذبي
بالتين اولهم
لداق وده

بالطبع Let $h(x) = f(x) - g(x)$

$[a, b]$

① $\because f(x)$ and $g(x)$ are continuous on $[a, b]$

$\because f(x)$ is continuous on $[a, b]$

② $h'(x) = f'(x) - g'(x)$

$\because f(x)$ and $g(x)$ are differentiable on (a, b)

$\because h(x)$ is differentiable on (a, b)

③ $h(a) = f(a) - g(a) = 0 \Rightarrow h(a) = h(b)$
 $h(b) = f(b) - g(b) = 0$

From ①②③

$\because$ Rolle's Thm is satisfied

then there is a number $c \in (a, b)$
such that

$$h'(c) = 0$$

$$f'(c) - g'(c) = 0$$

$$\boxed{f'(c) = g'(c)}$$

6) let f be differentiable on $[1, 3]$,

$$f(1) = 3, \quad f(3) = 1$$

Show that the graph of f admits (has) a tangent line

at $c \in (1, 3)$ parallel to the line of the equation $x + y - 4 = 0$

أثبت أن tangent line $\Leftarrow$ (مشتق) = (مشتق)₂

ولكن فاعني دالة أكثر أيب هذا المشتق الثاني

الخطتين الوحيدتين إلى بعض من فلالهم أيب المشتق هو

لو فاعني عند

M.V.T

Rolle

لأن دور الـ طرفي طرفي

$f(x)$

$[1, 3]$

$$x + y - 4 = 0$$

$$1 + y' = 0$$

$$y' = -1$$

$\because f$ is differentiable on $[1, 3]$ ①

$\because f$ is continuous on $[1, 3]$ ②

from ① ②

$\because$ M.V.T is satisfied

then there is a number c

$\in (1, 3)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$= \frac{f(3) - f(1)}{3 - 1} = \frac{1 - 3}{2} = \frac{-2}{2} = -1$$

$$\because f'(c) = y' = -1$$

$\because$ the tangent line of f is parallel to the line y

لا بد أن الدالة

Has at least one solution.

I.V.T

Exactly one solution.

Rolle's Theorem

+

I.V.T

Has at most one solution.

Rolle's Theorem

Use Rolle's theorem to show that for all $a \in \mathbb{R}$ the equation

$$\frac{3x}{2} + \sin x = a \text{ has at most one solution}$$

$$\frac{3x}{2} + \sin x - a = 0$$

$$\text{Let } h(x) = \frac{3}{2}x + \sin x - a = 0$$

Let x_1, x_2 are two different solutions of $h(x)$

① Domain of $h(x)$ is $= \mathbb{R}$ [Trigonometric function]
Polynomial of degree=1
 $\therefore h(x)$ is continuous on $[x_1, x_2]$

$$\begin{aligned} \text{② } h'(x) &= \frac{3}{2} + \cos x - 0 \\ &= \frac{3}{2} + \cos x \end{aligned}$$

$\therefore h(x)$ is differentiable on $\mathbb{R}$ [Trigonometric function]

$\therefore h(x)$ is differentiable on (x_1, x_2)

$$\begin{aligned} \text{③ } h(x_1) &= 0 \\ h(x_2) &= 0 \end{aligned} \rightarrow \therefore h(x_1) = h(x_2)$$

From ①②③ $\implies \therefore$ Rolle's Theorem is satisfied
then there is a number $c \in (x_1, x_2)$
such that $h'(c) = 0$

$$\frac{3}{2} + \cos c = 0$$

$$\begin{aligned} \cos c &= -\frac{3}{2} \\ \text{which is wrong} \\ -1 &\leq \cos c \leq 1 \end{aligned}$$

$\implies \therefore h$ has at most one solution.

(a) State Rolle's theorem

(b) Show that $f(x) = x^4 + 2x^2 - 3x + 1$ has at most one critical number

①

بم كتابه الترفيف

② $f'(x) = 4x^3 + 4x - 3$

Let $h(x) = f'(x)$

$$h(x) = 4x^3 + 4x - 3$$

Let x_1, x_2 are two different solutions of $h(x)$

① Domain of $h(x) = \mathbb{R}$ [Polynomial of degree 3
Cubic]

$\therefore h(x)$ is continuous on $[x_1, x_2]$

② $h'(x) = 12x^2 + 4$

$\therefore h(x)$ is differentiable on $\mathbb{R}$ [Polynomial of degree

$\therefore h(x)$ is differentiable on (x_1, x_2) Quadratic²]

③ $h(x_1) = 0$
 $h(x_2) = 0 \rightarrow \therefore h(x_1) = h(x_2)$

From ①②③ $\implies \therefore$ Rolle's Theorem is satisfied
then there is a number $c \in (x_1, x_2)$

such that $h'(c) = 0$

$$12c^2 + 4 = 0 \rightarrow c^2 = \frac{-4}{12} = -\frac{1}{3}$$

which is wrong

$\Rightarrow \therefore f(x)$ has at most one critical number

Use Rolle's theorem to show the graph of

$$f(x) = x^4 + x^3 - 3x^2 + x + 1$$

can not have more than one inflection point on $[0, 1]$

$$f(x) = x^4 + x^3 - 3x^2 + x + 1$$

$[0, 1]$

$$f'(x) = 4x^3 + 3x^2 - 6x + 1 \quad \rightarrow \quad f''(x) = 12x^2 + 6x - 6$$

Let $h(x) = f''(x)$

$$h(x) = 12x^2 + 6x - 6$$

Let x_1, x_2 are two different solutions of $h(x)$

① Domain of $h(x) = \mathbb{R}$ [Polynomial of degree 2]

$\therefore h(x)$ is continuous on $[x_1, x_2]$ Quadratic

② $h'(x) = 24x + 6$

$\therefore h(x)$ is differentiable on $\mathbb{R}$ [Polynomial of degree 1]
Linear

③ $h(x_1) = 0$
 $h(x_2) = 0 \rightarrow \therefore h(x_1) = h(x_2)$

From ① ② ③ $\implies \therefore$ Rolle's Theorem is satisfied
Then there is a number $c \in (x_1, x_2)$

such that $h'(c) = 0$
 $24c + 6 = 0 \rightarrow c = -\frac{6}{24} = -\frac{1}{4} \notin (0, 1)$

$\Rightarrow \therefore f$ Can Not have more than one Inflection Point.

$$\text{Let } f(x) = x^5 + x^3 + 20x - 10$$

Show that f has exactly one solution.

Let x_1 and x_2 are two different solutions of f .

$$f(x) = x^5 + x^3 + 20x - 10 \quad [x_1, x_2]$$

1) Domain of $f(x) = \mathbb{R}$

$\therefore f(x)$ is continuous on $[x_1, x_2]$

2) $f'(x) = 5x^4 + 3x^2 + 20$

$f(x)$ is differentiable on $\mathbb{R}$

$\therefore f(x)$ is differentiable on $[x_1, x_2]$

3) $f(x_1) = 0$
 $f(x_2) = 0 \longrightarrow \therefore f(x_1) = f(x_2)$

From ① ② ③ $\longrightarrow \therefore$ Rolle's Theorem is satisfied
then there is a number $c \in (x_1, x_2)$ such that

$$f'(c) = 0$$

$$5c^4 + 3c^2 + 20 = 0$$

$$5c^4 + 3c^2 = -20 \quad (\text{which is wrong})$$

⊕ ⊕ 7, 0

$\therefore f(x)$ has at most one solution $\longrightarrow \triangle 1$

$$f(x) = x^5 + x^3 + 20x - 10$$

$$1) f(0) = 0 + 0 + 0 - 10 = -10 < 0$$

$$2) f(1) = 1 + 1 + 20 - 10 = 12 > 0$$

From ① ② $\longrightarrow \therefore$ By I.V.T

Since $f(x)$ is continuous on $\mathbb{R}$

then $f(x)$ is continuous on $[0, 1]$

then there exist at least one $c \in (0, 1)$

such that $f(c) = 0 \longrightarrow \triangle 2$

From $\triangle 1$ $\triangle 2$

$\therefore f(x)$ has exactly one solution.

Show that the equation has exactly one real root.

$$x^3 + e^x = 0$$

Past exam

$$f(x) = x^3 + e^x \quad [x_1, x_2]$$

Let x_1 and x_2 are two different solutions of f .

1) Domain of $f(x) = \mathbb{R}$

$\therefore f(x)$ is continuous on $[x_1, x_2]$

2) $f'(x) = 3x^2 + e^x$

$f(x)$ is differentiable on $\mathbb{R}$

$\therefore f(x)$ is differentiable on (x_1, x_2)

3) $f(x_1) = 0$
 $f(x_2) = 0 \longrightarrow \therefore f(x_1) = f(x_2)$

From ① ② ③ $\longrightarrow \therefore$ Rolle's Theorem is satisfied

then there is a number $c \in (x_1, x_2)$ such that

$$f'(c) = 0$$

$$3c^2 + e^c = 0 \quad (\text{which is wrong})$$

$\nearrow 0 \quad \nearrow 0 \quad \nearrow 0$

$\therefore f(x)$ has at most one solution $\longrightarrow \triangle$

$$f(x) = x^3 + e^x$$

$$1) f(0) = 0 + e^0 = 1 > 0$$

$$2) f(-1) = -1 + e^{-1} = -1 + \frac{1}{e} < 0$$

From ① ② $\longrightarrow \therefore$ By I.V.T

Since $f(x)$ is continuous on $\mathbb{R}$ [x^3 : polynomial of $d=3$
 e^x : exponential function]

then $f(x)$ is continuous on $[-1, 0]$

then there exist at least one $c \in (-1, 0)$

such that $f(c) = 0 \longrightarrow \triangle 2$

From $\triangle 1$ $\triangle 2$

$\therefore f(x)$ has exactly one real root

استخدام Mean Value Theorem

لائحات المتباينات

عند وجود فترة

$[a, b]$



$$a < c < b$$

عند وجود

$$\sin c - \cos c$$

مقلوباتها
أو



$$-1 \leq \sin c \leq 1$$

$$-1 \leq \cos c \leq 1$$

$$0 \leq \sin c \leq 1$$

$$0 \leq \cos c \leq 1$$



Text book question



Use the Mean Value Theorem to prove the inequality

$$|\sin a - \sin b| \leq |a - b| \text{ for all } a \text{ and } b$$

$$f(x) = \sin x \quad [a, b]$$

1) Domain of $f(x) = \mathbb{R}$

$\therefore f(x)$ is continuous on $[a, b]$

2) $f'(x) = \cos x$

$f(x)$ is differentiable on $\mathbb{R}$

$\therefore f(x)$ is differentiable on (a, b)

From ①② $\longrightarrow \therefore$ Mean Value Theorem is satisfied
then there is $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\cos c = \frac{\sin b - \sin a}{b - a}$$

$$|\cos c| = \left| \frac{\sin b - \sin a}{b - a} \right| \longrightarrow (1)$$

$$|\cos c| \leq 1$$

$$\left| \frac{\sin b - \sin a}{b - a} \right| \leq 1$$

$$\frac{|\sin b - \sin a|}{|b - a|} \leq 1$$

$$\longrightarrow |\sin b - \sin a| \leq |b - a|$$

$$\therefore |\sin a - \sin b| \leq |a - b|$$

Use the Mean Value Theorem to show that if u and v are any real numbers, then

$$|\cos^7 u - \cos^7 v| \leq 7 |u - v|$$

$f(x) = \cos^7 x \quad [u, v]$

1) Domain of $f(x) = \mathbb{R}$ [Trigonometric function]

$\therefore f(x)$ is continuous on $[u, v]$

2) $f'(x) = 7 \cos^6 x (-\sin x) = -7 \cos^6 x \sin x$

$f(x)$ is differentiable on $\mathbb{R}$ [Trigonometric function]

$\therefore f$ is differentiable on $[u, v]$

From ① ② $\longrightarrow \therefore$ Mean Value Theorem is satisfied

then there exist at least one $c \in (u, v)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$-7 \cos^6 c \cdot \sin c = \frac{f(v) - f(u)}{v - u}$$

$$-7 \cos^6 c \sin c = \frac{\cos^7 v - \cos^7 u}{v - u}$$

$$|-7 \cos^6 c \sin c| = \left| \frac{\cos^7 v - \cos^7 u}{v - u} \right|$$

$$7 |\cos^6 c| |\sin c| = \left| \frac{\cos^7 v - \cos^7 u}{v - u} \right| \longrightarrow (1)$$

*7 $|\cos^6 c| |\sin c| \leq 1$

$7 |\cos^6 c| |\sin c| \leq 7 \longrightarrow (2)$

(1) in (2) $\frac{|\cos^7 v - \cos^7 u|}{|v - u|} \leq 7 \longrightarrow |\cos^7 v - \cos^7 u| \leq 7 |v - u|$
 $\therefore |\cos^7 u - \cos^7 v| \leq 7 |u - v|$

Use mean value theorem to show that if $0 < a < b$

then $\sqrt{b} - \sqrt{a} < \frac{b-a}{2\sqrt{a}}$

$f(x) = \sqrt{x}$ $[a, b]$

① Domain of $f(x) = [0, \infty)$ [Root function]

$\therefore f(x)$ is continuous on $[a, b]$

② $f'(x) = \frac{1}{2\sqrt{x}}$

f is differentiable on $(0, \infty)$ [Algebraic function]

$\therefore f$ is differentiable on (a, b)

From ① ② $\rightarrow \therefore$ Mean Value Theorem is satisfied
then there is a number $c \in (a, b)$

such that $f'(c) = \frac{f(b) - f(a)}{b-a}$

$\frac{1}{2\sqrt{c}} = \frac{\sqrt{b} - \sqrt{a}}{b-a}$

Now

$a < c < b$

$\sqrt{a} < \sqrt{c} < \sqrt{b}$

*2

$2\sqrt{a} < 2\sqrt{c} < 2\sqrt{b}$

$\frac{1}{2\sqrt{a}} > \frac{1}{2\sqrt{c}} > \frac{1}{2\sqrt{b}}$
 $\frac{1}{2\sqrt{a}} > \frac{\sqrt{b} - \sqrt{a}}{b-a} > \frac{1}{2\sqrt{b}}$
 $\frac{b-a}{2\sqrt{a}} > \sqrt{b} - \sqrt{a} < \frac{b-a}{2\sqrt{b}}$
 $\Rightarrow \therefore \frac{b-a}{2\sqrt{a}} > \sqrt{b} - \sqrt{a}$

Use the Mean Value Theorem to show that

$$\tan\left(\frac{3}{2}\right) - \tan\left(\frac{1}{2}\right) > 1$$

$$f(x) = \tan x, \quad \left[\frac{1}{2}, \frac{3}{2}\right]$$

1) Domain of $f(x) = \mathbb{R} \setminus \{\frac{\pi}{2} + n\pi\}$

$\therefore f(x)$ is continuous on $[\frac{1}{2}, \frac{3}{2}]$

2) $f'(x) = \sec^2 x$

$f(x)$ is differentiable on $\mathbb{R} \setminus \{\frac{\pi}{2} + n\pi\}$

$\therefore f(x)$ is differentiable on $(\frac{1}{2}, \frac{3}{2})$

From ①② $\longrightarrow \therefore$ Mean Value Theorem is satisfied

then there is a number $c \in (\frac{1}{2}, \frac{3}{2})$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\sec^2 c = \frac{f(\frac{3}{2}) - f(\frac{1}{2})}{\frac{3}{2} - \frac{1}{2}}$$

$$\sec^2 c = \frac{\tan(\frac{3}{2}) - \tan(\frac{1}{2})}{\frac{2}{2}}$$

$$\frac{1}{\cos^2 c} = \tan(\frac{3}{2}) - \tan(\frac{1}{2}) \longrightarrow (1)$$

$$\cos^2 c < 1 \quad c \in (\frac{1}{2}, \frac{3}{2})$$

$$\frac{1}{\cos^2 c} > 1 \longrightarrow (2)$$

(1) in (2)

$$\tan(\frac{3}{2}) - \tan(\frac{1}{2}) > 1$$



Text book question



Show that $\sin x < x$ if $0 < x < 2\pi$

$$f(x) = \sin x \quad [0, x]$$

1) Domain of $f(x) = \mathbb{R}$

* $f(x)$ is continuous on $[0, x]$

2) $f'(x) = \cos x$

$f(x)$ is differentiable on $\mathbb{R}$

$\therefore f(x)$ is differentiable on $(0, x)$

From ① ② $\longrightarrow \therefore$ Mean Value Theorem is satisfied
then there is a number $c \in (0, x)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\cos c = \frac{f(x) - f(0)}{x - 0}$$

$$\cos c = \frac{\sin x - 0}{x}$$

$$\cos c = \frac{\sin x}{x} \longrightarrow \textcircled{1}$$

$$\therefore -1 \leq \cos c \leq 1 \longrightarrow \textcircled{2}$$

① in ②

$$\frac{\sin x}{x} \leq 1 \longrightarrow \sin x \leq x$$

Use the Mean Value Theorem to show that

$$(1+x)^{\frac{2}{3}} < 1 + \frac{2}{3}x, \quad x > 0$$

Past exam

$$f(x) = (1+x)^{\frac{2}{3}} \quad [0, x]$$

1) Domain of $f(x) = \mathbb{R}$ [Algebraic function]

$\therefore f(x)$ is continuous on $[0, x]$

$$2) f'(x) = \frac{2}{3}(1+x)^{-\frac{1}{3}} = \frac{2}{3(1+x)^{\frac{1}{3}}}$$

$f(x)$ is differentiable on $\mathbb{R} \setminus \{-1\}$

$\therefore -1 \notin (0, x)$

$\therefore f(x)$ is differentiable on $(0, x)$

From ① ② $\longrightarrow \therefore$ Mean Value Theorem is satisfied

then there is a number $c \in (0, x)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\frac{2}{3(1+c)^{\frac{1}{3}}} = \frac{f(x) - f(0)}{x - 0}$$

$$\frac{2}{3(1+c)^{\frac{1}{3}}} = \frac{(1+x)^{\frac{2}{3}} - (1)^{\frac{2}{3}}}{x}$$

$$\frac{2}{3(1+c)^{\frac{1}{3}}} = \frac{(1+x)^{\frac{2}{3}} - 1}{x} \longrightarrow (1)$$

$$0 < c < x$$

$$1 < 1+c < 1+x$$

$$(1)^{\frac{1}{3}} < (1+c)^{\frac{1}{3}} < (1+x)^{\frac{1}{3}}$$

$$* \frac{2}{3} \cdot \frac{1}{1} > \frac{1}{(1+c)^{\frac{1}{3}}} > \frac{1}{(1+x)^{\frac{1}{3}}}$$

$$\frac{2}{3} > \frac{2}{3(1+c)^{\frac{1}{3}}} > \frac{2}{3(1+x)^{\frac{1}{3}}}$$

(1) in (2)

$$\frac{2}{3} > \frac{(1+x)^{\frac{2}{3}} - 1}{x}$$

$$*x \cdot \frac{2}{3} > (1+x)^{\frac{2}{3}} - 1$$

$$\frac{2}{3}x + 1 > (1+x)^{\frac{2}{3}} \longleftarrow (2)$$